

R-GROUNDBENCH: A Diagnostic Benchmark for R-Group Grounding in Markush Molecular Editing

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Abstract

Recent advances in AI for scientific discovery enable molecular understanding and design, yet reasoning over incomplete chemical representations remains unclear. Markush structures, which encode molecular families through variable R-group placeholders (R_1 , R_2 , X , etc.), are ubiquitous in pharmaceutical patents and require grounding across molecular, textual, and chemical information. However, existing molecule-language benchmarks focus on fully specified molecules, leaving R-group grounding largely unevaluated. We introduce R-GROUNDBENCH, a diagnostic benchmark built from real patent Markush structures, featuring a Multiple-Choice (VQA) track with controlled difficulty and modality splits, and an open-ended Generation track. Our results reveal a substantial gap between recognition and molecular grounding. While models achieve over 90% accuracy on Easy VQA, performance drops to 56–66% on Hard VQA when shortcuts are controlled. Chemical-domain VLMs also remain unreliable, achieving only 25.7–46.2% on Hard VQA despite domain-specific pre-training. Moreover, Generation Exact Match remains below 20% for most models and below 8% when visual input is required. These findings reveal that current AI systems lack reliable grounding and execution for Markush editing, highlighting challenges for AI-driven scientific discovery. Code and benchmark data are publicly available at  GitHub and Hugging Face.

1 Introduction

Foundation models have shown increasing potential for AI-assisted scientific discovery, yet their ability to understand structured scientific objects and perform precise domain-specific transformations remains unclear. Reliable scientific AI requires more than generating plausible outputs: models must interpret structured representations, connect information across different modalities, follow symbolic design constraints, and faithfully execute scientific transformations. These capabilities are essential for reliable scientific AI, but remain largely unexplored in existing evaluations.

The rapid progress of molecular AI has produced benchmarks for optical chemical structure recognition (Morin et al. 2023; Fang et al. 2025), molecule captioning and

text-guided generation (Fang et al. 2023), complete-molecule editing (Zhuang, Shen, and Sun 2025), and chemical reasoning (Hao et al. 2026). These benchmarks have significantly advanced the evaluation of molecular understanding and generation. However, they primarily focus on fully specified molecules, where the complete molecular structure is already available. Such settings allow models to operate on explicit molecular graphs without resolving latent chemical variables. They therefore do not examine whether models can ground abstract molecular placeholders to concrete chemical structures before performing molecular instantiation. As a result, it remains unclear whether current general-purpose language/vision-language models and chemical-domain models possess genuine molecular grounding ability or rely on superficial structural shortcuts.

We investigate this missing capability through Markush structures (Markush 1924), a representation widely used in pharmaceutical patents to describe families of related molecules using variable placeholders (e.g., R_1 , R_2 , and X). Unlike conventional molecular representations that specify a single compound, Markush structures require models to associate abstract substituent variables with their corresponding chemical realizations. We define this capability as *R-group grounding*: the ability to localize variable positions, resolve intended substitutions from instructions, and instantiate the corresponding molecular structures. Although fundamental to reliable molecular editing, this capability has not been systematically evaluated by existing benchmarks.

To address this gap, we introduce R-GROUNDBENCH, a diagnostic benchmark for multimodal R-group grounded Markush molecular editing. An overview of the benchmark motivation, evaluation tracks, and core findings is presented in Figure 1. R-GROUNDBENCH is constructed from real patent and literature Markush structures sourced from MolParser-7M (Fang et al. 2025), containing aligned molecular images, E-SMILES representations, natural-language editing instructions, edited structures, and molecular annotations.

R-GROUNDBENCH evaluates models through two complementary settings. The **Multiple-Choice (VQA)** track measures whether models can identify the correct edited structure

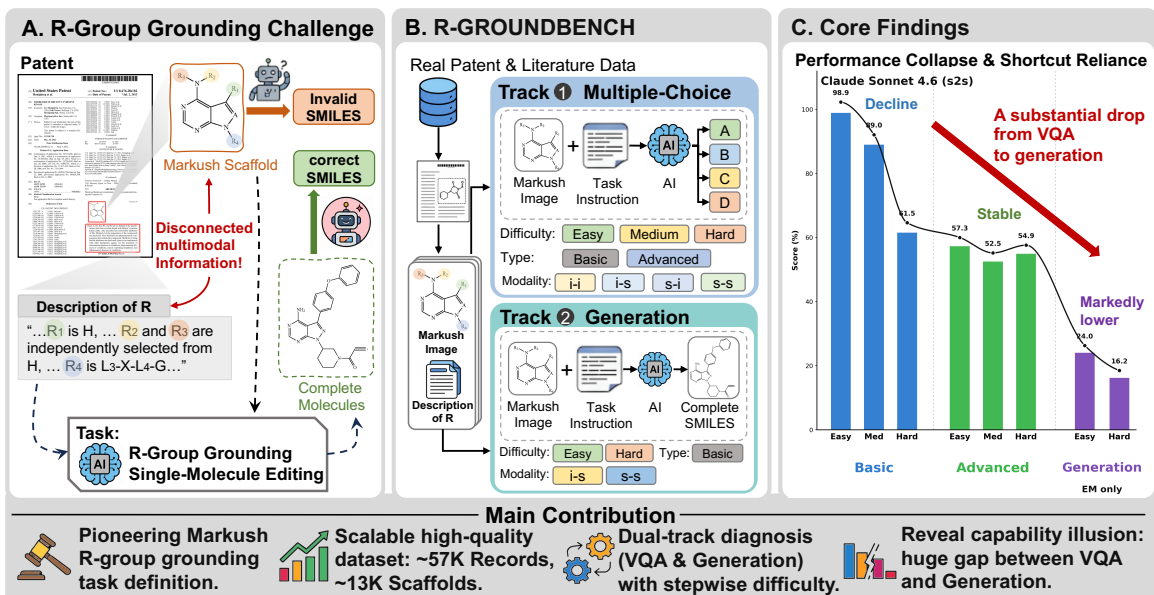


Figure 1: Overview of R-GROUNDBENCH. **A**: R-group grounding requires aligning disconnected molecular, textual, and symbolic information. **B**: R-GROUNDBENCH evaluates this capability through complementary VQA and Generation tracks with controlled difficulty and modality settings. **C**: Results reveal a gap between candidate recognition and autonomous molecular instantiation, exposing shortcut reliance in current models.

among candidates under controlled difficulty levels, while the **Generation** track requires models to directly instantiate the edited molecule without candidate options. Together, these settings distinguish candidate-level recognition from end-to-end molecular editing ability. The benchmark further supports four image/SMILES input-output modalities, enabling systematic analysis of representation effects.

Our main contributions are:

- We introduce R-group grounded Markush molecular editing as a diagnostic task for evaluating molecular grounding ability in foundation models, connecting multimodal reasoning with AI for scientific discovery.
- We construct R-GROUNDBENCH from the MolParser-7M sft_real subset, filtering 46,727 Markush-containing patent records into 14,394 clean single-scaffold samples, resulting in 56,500 instruction records across 12,962 unique Markush scaffolds.
- We benchmark general-purpose LLMs, VLMs, and chemical-domain models on R-GROUNDBENCH, revealing a large gap between candidate recognition and autonomous molecular instantiation: models perform strongly on simple VQA but degrade substantially on hard grounding and generation.

2 Related Work

Molecular Structure and Patent Analysis. Optical chemical structure recognition (OCSR) converts molecular images into machine-readable representations such as SMILES (Valko and Johnson 2009; Filippov and Nicklaus 2009). Recent methods, including MolGrapher (Morin et al. 2023), DECIMER (Rajan, Zielesny, and Steinbeck 2023), and MolParser (Fang et al. 2025), improve molecular struc-

Benchmark	Markush	R-grnd.	Editing	Multimed.	Shortcut
MolGrapher	×	×	×	img	×
DECIMER	×	×	×	img	×
MolParser	✓	×	×	img	×
MarkushGrapher	✓	×	×	img+txt	×
MolLangBench	×	×	✓	img+txt	×
Mol-Instructions	×	×	✓	txt	×
MolEditRL	×	×	✓	txt	×
ChemCoTBench	×	×	✓	txt	×
R-GROUNDBENCH (Ours)	✓	✓	✓	img+txt	✓

Table 1: Comparison of related benchmarks. Markush: supports Markush structures; R-grnd.: evaluates R-group grounding; Editing: tests molecular editing; Multimod.: input modalities; Shortcut: diagnoses shortcut reasoning.

ture extraction from images. In particular, MolParser introduces E-SMILES to represent Markush structures by preserving R-group placeholders alongside molecular scaffolds, enabling machine-readable representations of patent-defined molecule families. For pharmaceutical patent analysis, PatentAgent (Wang et al. 2024) explores agent-based document understanding, while MarkushGrapher (Morin et al. 2025) extracts backbone graphs and substituent tables from patent documents. However, these approaches focus on molecular extraction or patent-level analysis, whereas R-GROUNDBENCH evaluates the subsequent capability of grounding R-group placeholders and performing instruction-guided molecular editing.

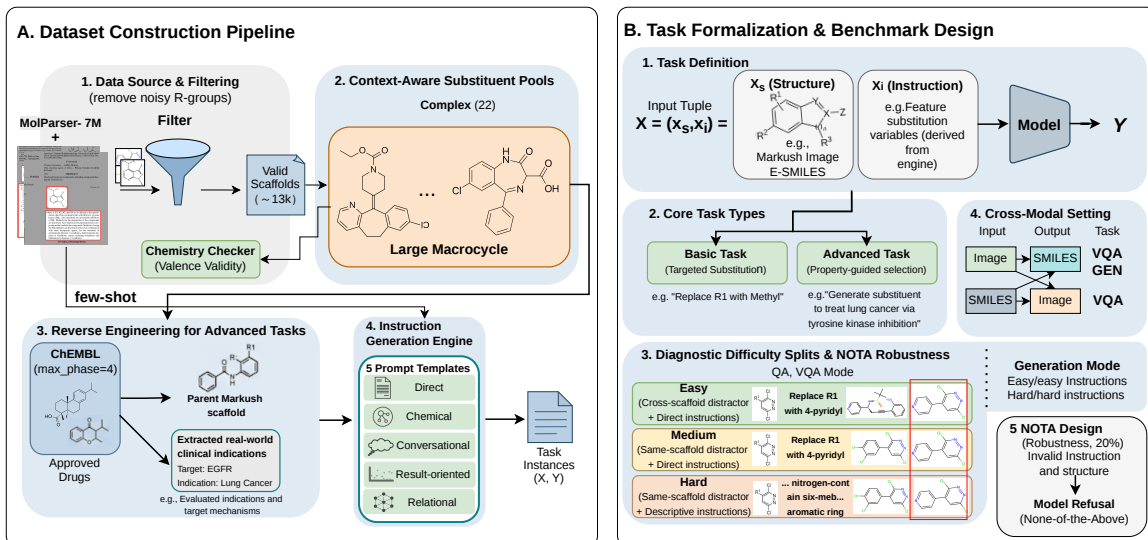


Figure 2: Overview of R-GROUNDBENCH construction and task formulation. **A**: Dataset construction from real Markush structures with filtering, annotation, and instruction generation. **B**: Dual-track evaluation design, including VQA with controlled difficulty/modality settings and Generation without candidate options.

Molecule-Language Benchmarks. MolLangBench (Cai et al. 2026) covers language-prompted molecular recognition, editing, and generation. Mol-Instructions (Fang et al. 2023) provides data for captioning, generation, and property prediction. MolEditRL (Zhuang, Shen, and Sun 2025) edit complete molecules via SMILES instructions. ChemCoTBench (Hao et al. 2026) uses modular reasoning for molecular transformations. Unlike prior work on complete, fixed molecules, R-GROUNDBENCH targets variable Markush scaffolds, requiring grounding of placeholder symbols before instantiation, mirroring real patent workflows. Table 1 summarizes the comparison between R-GroundBench and existing molecular AI benchmarks.

Shortcut Reasoning in Multimodal Evaluation. VQA models may exploit language biases (Goyal et al. 2017) or visual statistics (Agrawal et al. 2018; He et al. 2020) instead of genuine understanding. In molecular VQA, similar shortcuts can arise from matching substituent names rather than grounding molecular structures. R-GROUNDBENCH mitigates this issue through Easy/Medium/Hard splits, progressively removing such shortcuts via scaffold-level and instruction-level constraints. Beyond shortcut control, R-GROUNDBENCH addresses a limitation of complete-molecule editing benchmarks (Zhuang, Shen, and Sun 2025): Markush editing requires both placeholder localization and substituent instantiation from natural-language instructions. Errors in localization directly lead to incorrect molecular edits, making these two capabilities essential for reliable R-group grounding.

3 R-GROUNDBENCH

3.1 Data Construction

R-GroundBench is constructed through a three-stage pipeline: collecting and filtering real Markush structures, generating grounded R-group editing instances, and enriching samples with molecular annotations for advanced reasoning tasks. The resulting benchmark provides aligned

molecular images, E-SMILES representations, editing instructions, edited structures, and property annotations. The overall benchmark construction pipeline and task formulation are illustrated in Figure 2.

Step 1: Collecting and Filtering Markush Structures. We start from the `sft_real` subset of MolParser-7M (Fang et al. 2025), containing 91,166 molecular image–E-SMILES pairs extracted from patent and literature documents. Validity filters remove ambiguous structures and invalid representations, yielding 14,394 clean single-scaffold Markush structures. Multi-scaffold structures are excluded due to ambiguity in R-group localization.

Step 2: Constructing R-group Editing Instances. For each valid Markush structure, we parse R-group positions from E-SMILES, assign context-aware substituents, validate generated molecules with RDKit, and render 300×300 PNG images. We further construct natural-language editing instructions from real patent claim templates, ensuring that the benchmark reflects practical molecular editing scenarios. This process produces 55,982 unique editing operations across 12,962 scaffolds and 56,500 instruction records. Detailed filtering criteria, substituent construction, and instruction generation procedures are provided in Appendix.

Step 3: Adding Molecular Annotations. To support Advanced tasks, we enrich each editing instance with molecular properties. Six physicochemical descriptors are computed via RDKit (MW, logP, TPSA, HBD, HBA, and rotatable bonds). Biological annotations are obtained by reverse-engineering Markush structures from approved drugs in ChEMBL (Gaulton et al. 2012), linking molecules to bioactivity pIC_{50} , indication, and mechanism of action records. These annotations enable property-aware reasoning evaluation in the Advanced task type. The key statistics of R-GROUNDBENCH are summarized in Appendix.

3.2 Task Formulation

Definition and Two Tracks Let M denote a Markush representation, either a molecular image M_{img} or an E-SMILES string M_{smi} . E-SMILES encodes a Markush structure as a backbone SMILES (with R-group attachment points marked as $*$), followed by a `<sep>` delimiter and XML-like tokens that map each attachment point to its R-group label:

```
N1=CC(*)=C(*)N=C1*<sep>
<a>3:R[1]</a><a>5:R[2]</a><a>8:X</a>
```

Here atoms 3, 5, and 8 in the backbone carry placeholders R[1], R[2], and X, respectively. M_{img} and M_{smi} represent two modalities of the same Markush structure, and our benchmark evaluates models under both settings.

R-GROUNDBENCH evaluates R-group grounding through two complementary tracks that probe different stages of molecular instantiation. The Multiple-Choice (VQA) track evaluates candidate-level grounding by providing four candidate molecules and asking the model to select C^* from $\mathcal{C} = \{C_1, \dots, C_4\}$. In contrast, the Generation track removes candidate options and requires models to directly construct the edited molecule as a SMILES string given M and instruction I . Together, the two tracks distinguish candidate recognition from end-to-end molecular instantiation.

Multiple-Choice (VQA) Track The VQA track evaluates candidate-level R-group grounding through two task types and three difficulty levels. Basic and Advanced tasks examine different reasoning requirements after grounding, while Easy/Medium/Hard splits remove shortcut cues through controlled distractors and instruction design.

Task types and difficulty splits. In the Basic task, I specifies the desired substituent modification (e.g., "Replace R₁ with 4-chlorophenyl"). The model must localize the placeholder in M , ground it to the intended substituent, and select the candidate C^* with the correct structural edit. Basic tasks evaluate placeholder grounding and candidate instantiation without property reasoning.

In the Advanced task, I asks property-guided selection questions (e.g., "Which R-group is most hydrophobic?"). Property questions cover three categories: Physicochemical (25%, RDKit), Biological activity (45%, ChEMBL potency), and Indication/Mechanism (30%, ChEMBL approved drugs). Advanced tasks evaluate property-aware selection after grounding rather than property prediction.

We design three difficulty levels that progressively reduce shortcut cues by controlling distractor and instruction style. Easy uses cross-scaffold distractors with direct substituent names, allowing lexical matching shortcuts; Medium uses same-scaffold distractors with direct names to remove scaffold-level cues; Hard uses same-scaffold distractors with chemically descriptive instructions, requiring models to infer the intended substituent beyond lexical matching. Importantly, Hard is not designed to measure general language complexity; instead, it evaluates whether models can perform description-to-structure grounding after removing associations between substituent names and candidate structures.

Input-output modalities. To disentangle visual perception, symbolic manipulation, and cross-modal alignment, the

Mode	Input	Candidates	Tests
i2i	Markush image	mol. images	Visual grounding
i2s	Markush image	SMILES options	Image $\rightarrow$ symbol
s2i	E-SMILES	mol. images	Symbol $\rightarrow$ visual
s2s	E-SMILES	SMILES options	Symbolic editing

Table 2: The four input-output modality settings in R-GROUNDBENCH. Notation: $i2i$ = image in, image candidates; $i2s$ = image in, SMILES candidates; $s2i$ = E-SMILES in, image candidates; $s2s$ = E-SMILES in, SMILES candidates.

VQA track covers four input-output modality combinations that shown in Table 2. Text-only LLMs are evaluated solely under the s2s setting; VLMs are evaluated across all four settings, enabling us to attribute performance differences to visual perception failures, symbolic reasoning failures, or cross-modal alignment failures.

NOTA robustness. To test candidate rejection, 20% of Basic Easy/Medium questions include a NOTA option: 5% where NOTA is a distractor (correct among A/B/C); 10% where all candidates contain structural errors (NOTA correct); and 5% where the instruction references a non-existent R-group (NOTA correct). No NOTA questions appear in Hard splits, as Hard focuses on precise R-group grounding among valid rather than invalid candidates.

Generation Track The Generation track evaluates end-to-end molecular instantiation by requiring models to directly generate edited SMILES without candidate options. Unlike VQA, which measures candidate selection under controlled settings, Generation evaluates whether models can autonomously ground R-groups, realize molecular edits, and produce valid molecular representations.

Task design. The task requires models to complete the full editing pipeline, including R-group grounding, molecular realization, and SMILES generation. Although SMILES is the output representation, evaluation focuses on whether models preserve the Markush scaffold and perform instruction-grounded R-group substitution correctly.

We restrict Generation to Basic tasks because they directly evaluate molecular instantiation after substituent grounding. In contrast, Advanced tasks further involve property-aware reasoning, and introducing Generation in this setting would entangle failures in property understanding with failures in molecular realization. Generation therefore serves as a complementary evaluation setting following the grounding progression established in VQA.

Difficulty is controlled by instruction style: Easy uses direct substituent names, while Hard uses chemically descriptive instructions that require inference beyond explicit names. We evaluate two input modalities: s2s and i2s.

4 Experiments and Results

4.1 Experimental Setup

Models. We evaluate three categories of models: general-purpose Large Language Models, general-purpose vision-language models, and chemical-domain models specialized

Model Type	Model	Modality	Basic				Advanced				Δ_{EH}		Avg.
			Easy	Med	Hard	Avg.	Easy	Med	Hard	Avg.	Basic	Adv	
General VLMs	Qwen3-VL-8B (Qwen Team 2025b)	i2i	25.5	24.9	25.1	25.2	21.3	20.4	21.1	20.9	0.4	0.2	23.0
		i2s	93.6	74.8	55.7	74.7	55.2	54.7	55.7	55.2	37.9	-0.5	64.9
		s2i	78.4	68.7	49.6	65.6	37.7	38.2	37.2	37.7	28.8	0.5	51.6
		s2s	93.4	72.7	55.9	74.0	49.4	48.7	49.2	49.1	37.5	0.2	61.6
	Qwen3-VL-32B (Qwen Team 2025b)	i2i	<u>98.1</u>	90.2	55.5	81.3	44.6	44.2	47.0	45.3	42.6	-2.4	63.3
		i2s	<u>97.7</u>	91.3	60.9	83.3	<u>60.4</u>	<u>60.3</u>	61.0	<u>60.6</u>	36.8	-0.6	71.9
		s2i	95.2	90.6	55.6	80.5	46.4	47.2	44.0	45.9	39.6	2.4	63.2
		s2s	97.5	91.4	61.9	83.6	50.3	48.9	49.1	49.4	35.6	1.2	66.5
	Claude Sonnet 4.6(Anthropic 2026)	i2i	98.5	94.6	61.6	84.9	48.4	46.4	51.6	48.8	36.9	-3.2	66.8
		i2s	98.3	<u>94.8</u>	63.8	<u>85.6</u>	59.2	52.5	54.3	55.3	34.5	4.9	70.5
		s2i	97.9	86.0	59.7	81.2	53.3	50.8	52.4	52.2	38.2	0.9	66.7
		s2s	98.9	89.0	61.5	83.1	<u>57.3</u>	<u>52.5</u>	54.9	54.9	37.4	2.4	69.0
	GPT-4o (OpenAI 2024)	i2i	97.5	83.8	58.2	79.8	43.9	43.9	42.9	43.6	39.3	1.0	61.7
		i2s	96.8	90.6	60.6	82.7	57.3	58.2	52.1	55.9	36.2	5.2	69.3
		s2i	<u>97.5</u>	83.1	57.7	79.4	47.3	49.2	47.6	48.0	39.8	-0.3	63.7
		s2s	97.7	89.8	61.1	82.9	35.2	37.9	42.7	38.6	36.6	-7.5	60.7
	GPT-4.1 (OpenAI 2025a)	i2i	97.8	87.3	60.4	81.8	53.5	50.8	50.5	51.6	37.4	3.0	66.7
		i2s	97.5	90.4	62.8	83.6	59.0	60.7	<u>54.1</u>	57.9	34.7	4.9	70.8
		s2i	96.4	84.9	60.3	80.5	53.3	53.5	<u>50.2</u>	52.3	36.1	3.1	66.4
		s2s	97.3	90.5	61.9	83.2	39.1	41.6	42.1	40.9	35.4	-3.0	62.1
	GPT-5.5 (OpenAI 2025b)	i2i	96.8	95.5	64.3	85.5	<u>54.9</u>	<u>57.7</u>	<u>50.5</u>	<u>54.4</u>	32.5	4.4	<u>70.0</u>
		i2s	<u>97.9</u>	98.5	66.1	87.5	60.3	59.5	58.4	59.4	31.8	1.9	73.5
		s2i	<u>95.3</u>	94.0	64.4	84.6	<u>54.9</u>	<u>55.4</u>	<u>55.8</u>	<u>55.4</u>	30.9	-0.9	<u>70.0</u>
		s2s	<u>98.3</u>	98.5	<u>66.1</u>	87.6	58.8	48.1	53.8	53.6	32.2	5.0	70.6
	Gemini 2.5 Flash (Google DeepMind 2025)	i2i	95.3	93.8	62.6	83.9	59.8	58.2	62.2	60.1	32.7	-2.4	72.0
		i2s	97.5	<u>94.8</u>	<u>65.4</u>	<u>85.9</u>	62.5	60.3	<u>60.1</u>	61.0	32.1	2.4	<u>73.4</u>
		s2i	96.8	<u>93.4</u>	62.0	<u>84.1</u>	61.4	58.7	61.4	60.5	34.8	0.0	72.3
		s2s	98.1	<u>95.1</u>	63.8	<u>85.7</u>	54.7	53.5	<u>52.5</u>	<u>53.6</u>	34.3	2.2	<u>69.6</u>
	Llama 4 Maverick (Meta AI 2025)	i2i	92.7	86.2	59.5	79.5	35.2	36.0	33.3	34.8	33.2	1.9	57.2
		i2s	93.6	82.0	62.6	79.4	56.3	54.4	55.8	55.5	31.0	0.5	67.5
		s2i	89.7	81.2	59.9	76.9	36.0	38.2	37.5	37.2	29.8	-1.5	57.1
		s2s	90.6	82.3	63.7	78.9	33.6	36.3	37.7	35.9	26.9	-4.1	57.4
General LLMs	DeepSeek-V4-Pro (DeepSeek-AI 2026)	s2s	97.3	98.1	68.7	88.0	52.5	55.1	54.6	54.1	28.6	13.4	71.0
	DeepSeek-R1 (DeepSeek-AI 2025)	s2s	94.0	95.0	62.6	83.9	47.6	<u>54.7</u>	42.0	48.1	31.4	5.6	66.0
	LLaMA-3.3-70B-Instruct (Meta AI 2024)	s2s	96.7	79.6	60.4	78.9	42.6	42.9	44.2	43.2	36.3	-1.6	61.1
	o3 (OpenAI 2025c)	s2s	97.7	97.7	69.3	88.2	44.5	43.1	<u>52.5</u>	46.7	28.4	-8.0	67.5
	Qwen3-Max-2025-09-23 (Qwen Team 2025a)	s2s	98.6	<u>95.5</u>	64.2	86.1	<u>47.9</u>	49.2	<u>50.9</u>	<u>49.3</u>	34.4	-3.0	<u>67.7</u>
Chemical-domain Models	ChemVLM-8B (Li et al. 2025)	i2i	34.2	36.0	28.9	33.0	23.7	21.8	22.9	22.8	11.3	0.8	27.9
		i2s	96.8	85.9	56.7	79.8	63.0	63.4	63.8	63.4	33.0	-0.4	71.6
		s2i	62.2	53.9	40.2	52.1	35.8	38.5	37.4	37.2	24.8	-0.5	44.7
		s2s	96.3	84.2	58.8	79.7	61.2	60.4	61.5	61.0	34.8	0.7	70.4
	MolVL-7B (Fan et al. 2025)	i2i	22.0	20.5	25.5	22.7	22.1	29.4	24.0	25.2	-2.0	-3.9	23.9
		i2s	20.5	19.6	25.7	21.9	22.0	29.7	23.9	25.2	-3.4	-4.0	23.6
		s2i	21.7	19.5	25.2	22.1	21.9	30.0	23.8	25.2	-2.1	-3.9	23.7
		s2s	22.5	22.8	25.7	23.7	21.9	29.8	23.8	25.2	-1.3	-4.0	24.4

Table 3: Full VQA results on R-GROUNDBENCH (% accuracy). Columns: Basic/Advanced accuracy by difficulty (Easy/Med/Hard), their means, Δ_{EH} (Easy minus Hard), and overall mean across six splits. Rows: each model $\times$ modality (i2i, i2s, s2i, s2s); LLMs on s2s only. **bold**: best; underline: second best among models within each modality and model-type block. Chemical-domain models are not marked because model group contains fewer than three models.

in molecular understanding or structure processing. The latter includes chemical VLMs (ChemVLM-8B and MolVL-7B) and the Markush-specific model MarkushGrapher-2. Model details and citations are listed in Table 3.

Evaluation Metrics. For the VQA track, we report split-wise accuracy for Basic tasks and property-type accuracy for Advanced tasks. We define $\Delta_{EH} = \text{Acc}_{\text{Easy}} - \text{Acc}_{\text{Hard}}$ to quantify the performance drop after shortcut removal, and

report NOTA accuracy on Easy/Medium Basic splits. For the Generation track, we report three metrics with increasing tolerance to generation errors. Exact Match (EM) is the primary metric after canonicalizing predictions and ground truth. Scaffold Match measures backbone preservation using Maximum Common Substructure (MCS), requiring $\geq 80\%$ scaffold heavy-atom coverage. Tanimoto Similarity provides a continuous structural similarity measure when EM fails.

Model	Easy			Hard		
	EM	Scaff.	Tani.	EM	Scaff.	Tani.
<i>VLMs — s2s</i>						
Qwen3-VL-8B	4.2	49.2	44.0	3.0	52.2	45.6
Qwen3-VL-32B	12.8	38.0	57.7	10.0	41.4	52.9
Claude Sonnet 4.6	<u>24.0</u>	31.2	88.6	<u>16.2</u>	23.0	87.5
GPT-4o	19.4	53.0	59.9	10.4	50.8	54.0
GPT-4.1	15.4	48.2	57.3	9.8	51.0	53.1
GPT-5.5	44.6	62.2	78.4	38.4	64.0	76.7
Gemini 2.5 Flash	20.4	<u>55.8</u>	<u>62.5</u>	20.8	<u>56.4</u>	<u>62.6</u>
Llama 4 Maverick	12.2	48.6	52.0	9.0	<u>51.2</u>	48.9
<i>VLMs — i2s</i>						
Qwen3-VL-8B	1.2	25.0	22.3	0.1	25.6	23.3
Qwen3-VL-32B	2.6	31.0	34.1	2.6	33.0	32.6
Claude Sonnet 4.6	<u>10.2</u>	23.8	66.5	7.8	17.4	72.9
GPT-4o	4.4	31.4	36.5	1.8	23.2	34.4
GPT-4.1	4.2	31.4	33.8	1.8	27.6	35.4
GPT-5.5	16.2	51.8	<u>59.8</u>	17.0	54.0	<u>61.3</u>
Gemini 2.5 Flash	8.2	<u>42.2</u>	<u>51.1</u>	5.6	40.4	48.2
Llama 4 Maverick	7.0	<u>45.0</u>	43.1	5.2	<u>44.4</u>	40.8
<i>LLMs — s2s</i>						
DeepSeek-V4-Pro	50.7	72.7	82.6	29.7	<u>54.6</u>	71.7
DeepSeek-R1	24.5	45.5	<u>73.1</u>	20.3	45.9	64.0
LLaMA-3.3-70B	4.8	35.6	47.1	1.6	38.2	39.5
o3	<u>37.1</u>	<u>64.7</u>	76.6	<u>28.7</u>	64.1	<u>72.2</u>
Qwen3-Max	15.6	50.0	58.1	13.8	48.6	56.2
<i>Chemical-domain Models — i2s</i>						
MarkushGrapher-2	5.4	14.8	68.6	0.0	0.0	0.0
ChemVLM-8B	0.6	37.8	32.8	0.6	41.6	30.9
MolVL-7B	0.0	28.4	18.0	0.0	24.6	19.1
<i>Chemical-domain Models — s2s</i>						
ChemVLM-8B	2.8	34.0	41.4	1.8	36.2	36.9
MolVL-7B	0.0	18.2	19.5	0.0	15.2	16.8

Table 4: Generation track results (%). EM = Exact Match, Scaff. = Scaffold Match, Tani. = Tanimoto Similarity. Chemical-domain models are not marked.

Baselines. For VQA, random selection among four candidates provides a 25% chance-level baseline. For Generation, we introduce a rule-based *template oracle* as a non-learning reference to quantify performance achievable through explicit substituent-name matching. The oracle extracts substituent names from Easy instructions, retrieves corresponding SMILES from a lookup table, and performs RDKit-based substitution without learned representation or reasoning. The oracle achieves 88.0% EM and 0.969 Tanimoto on Easy Generation, confirming solvability. For Hard Generation, chemically descriptive instructions contain no explicit substituent names that can be directly matched (parse rate = 0%), resulting in 0.0% EM for the oracle. Although selected molecular editing cases are verified by chemistry experts during benchmark construction, we do not provide a large-scale human performance baseline due to the difficulty of recruiting sufficient qualified evaluators with expertise in Markush structures and molecular editing. This contrast demonstrates that Hard Generation removes name-matching shortcuts and requires instruction-grounded molecular instantiation rather than explicit substituent retrieval.

4.2 VQA Results and Analysis

Table 3 summarizes VQA performance across models, modalities, and difficulty levels. Three findings emerge:

- **Finding 1: Easy accuracy is high but shortcut-sensitive.** Most VLMs exceed 90% accuracy on Easy-Basic, but performance drops to 50–66% on Hard-Basic with same-scaffold distractors and shortcut-resistant instructions. For example, average s2i performance across eight VLMs decreases from 93.4% to 58.6%, suggesting that Easy results mainly reflect candidate discrimination rather than R-group grounding. Under s2s Hard-Basic, LLMs remain competitive with VLMs (64.1% vs. 61.4%), with o3 achieving the highest score (69.3%), showing that reasoning can partially compensate for missing visual input.
- **Finding 2: Representation alignment dominates modality effects.** Across four input-output modalities, Hard-Basic performance follows: i2s (62.2) $\approx$ s2s (62.0) > i2i (60.3) > s2i (58.6). The comparable performance between i2s and s2s suggests that visual input itself is not the primary bottleneck. Instead, the main challenge lies in aligning molecular structures, symbolic representations, and editing operations, particularly when models must translate between symbolic and visual representations.
- **Finding 3: Chemical-domain models do not guarantee R-group grounding ability.** Chemical-domain VLMs provide a test of whether molecular pretraining transfers to R-group grounding. Although they are trained on chemical representations, they achieve substantially lower performance than general-purpose VLMs on Basic VQA. This indicates that molecular representation learning alone does not guarantee reliable grounding of substituent instructions to candidate structures.

NOTA Robustness. We evaluate candidate rejection using NOTA questions (details in Appendix). Results show that invalid candidate rejection remains challenging. For example, GPT-4.1 achieves 98.8% on NOTA-as-distractor but only 76.2% when structural errors require selecting NOTA, suggesting that reliable validation requires explicit structural checking rather than answer-selection heuristics.

Property-aware Chemical Reasoning. We analyze Advanced VQA by property type. Bioactivity questions are the most challenging, with VLMs and LLMs achieving only 26–28%, close to the 25% random baseline, likely due to the difficulty of inferring experimental activity without assay-specific or SAR context. Physicochemical and indication questions are easier due to computable properties and memorized knowledge, showing that property-aware reasoning remains challenging beyond basic R-group substitution.

4.3 Generation Results and Analysis

Table 4 reveals a substantial gap between VQA selection and instruction-conditioned molecular generation. While VQA provides explicit candidates that reduce the search space, Generation requires autonomous R-group grounding, molecular instantiation, and SMILES construction. Although previous molecular instruction-following studies have shown

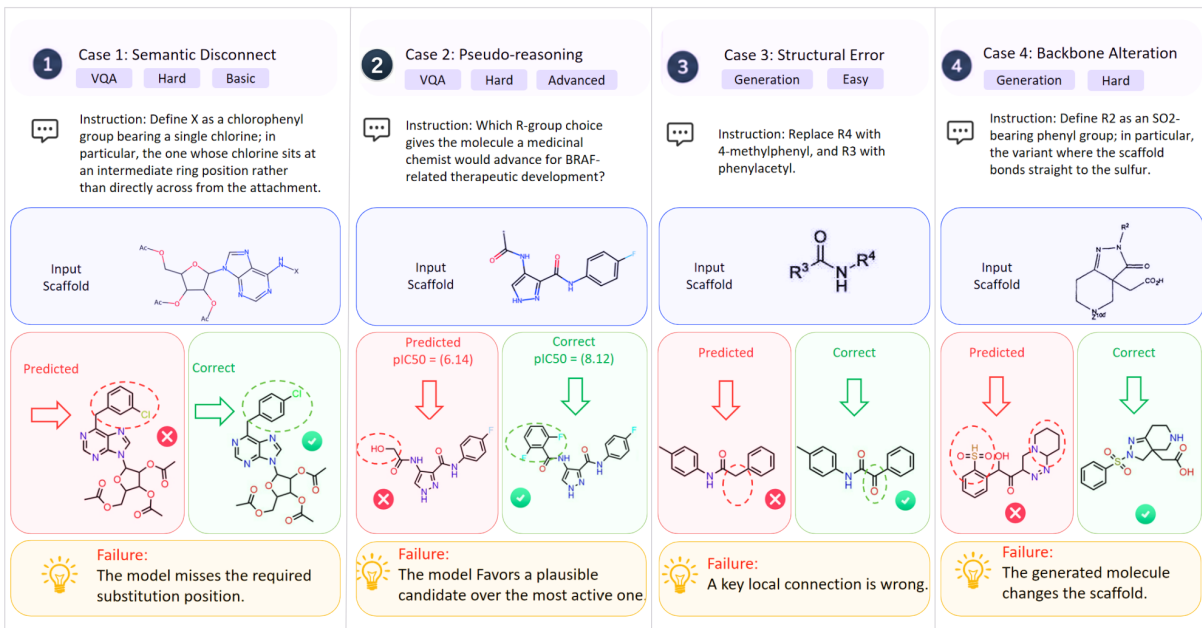


Figure 3: Four representative failure cases of GPT-4.1 on R-GROUNDBENCH.

that LLMs can generate valid molecular representations (Fang et al. 2023), R-GROUNDBENCH Generation additionally requires resolving which R-group to modify and how to instantiate the corresponding substituent. The average VLM Exact Match decreases from 59.7% in Hard-Basic VQA to 14.7% (s2s) and 5.2% (i2s) in Hard Generation.

Recognition does not translate into molecular instantiation. The large VQA-to-Generation gap indicates that candidate-level recognition does not directly translate into executable editing ability. When candidates are provided, models can often identify plausible structures; however, they struggle to independently resolve R-group assignments and construct the final molecular representation.

Symbolic input remains easier than visual generation. Across VLMs, s2s consistently outperforms image-based generation: average EM decreases from 19.1% to 6.8% on Easy and from 14.7% to 5.2% on Hard when moving from s2s to i2s. This suggests that visual input introduces additional alignment challenges, while the core difficulty remains precise structural modification.

Chemical-domain models do not guarantee instruction-grounded R-group generation. MarkushGrapher-2 (Strohmeier et al. 2026), despite being specifically designed for Markush structure reconstruction, only addresses part of the required pipeline and does not perform instruction-conditioned R-group substitution. It therefore achieves limited performance when required to instantiate R-group instructions into complete molecules. These results suggest that task-specific molecular generation capability does not necessarily imply robust instruction-grounded molecular editing.

4.4 Failure Case Studies

Four representative failure cases are shown in Figure 3.

Case 1: Semantic Disconnect. The model identifies the correct chlorophenyl category but selects the wrong substitution position, confusing an intermediate-position chlorine with a para substitution. This reveals a gap between linguistic interpretation and fine-grained R-group grounding.

Case 2: Pseudo-reasoning. In a BRAF-related bioactivity task, the model selects candidate A (pIC₅₀=6.14) instead of candidate C (pIC₅₀=8.52), despite a plausible rationale. This indicates that fluent explanations do not necessarily reflect reliable property-aware comparison.

Case 3: Structural Error. The model generates a valid SMILES with moderate similarity to the target, but introduces incorrect local connectivity and carbonyl patterns. The generated molecule appears chemically plausible and preserves part of the scaffold, yet fails to realize the requested R-group substitution. This demonstrates the gap between producing plausible molecules and performing precise molecular instantiation.

Case 4: Backbone Alteration. The model modifies the molecular backbone instead of applying the requested R-group substitution, resulting in a structurally different molecule. This highlights the challenge of preserving the Markush scaffold during autonomous editing.

5 Conclusion

We introduced R-GROUNDBENCH, a diagnostic benchmark for evaluating R-group grounding in Markush molecular editing. Our results reveal that current molecular AI capabilities are often overestimated when evaluation only requires candidate recognition. Although models achieve strong Easy VQA performance, they degrade under harder grounding conditions and instruction-conditioned Generation, exposing a gap between recognizing plausible structures and autonomously instantiating valid molecules.

Beyond a dataset, R-GROUNDBENCH provides a diagnostic framework for separating recognition, grounding, and execu-

tion through complementary evaluations, including the Δ_{EH} gap and VQA→Generation drop. These analyses show that current molecular AI systems still rely on superficial cues and lack structure-aware grounding for molecular editing.

Our findings further identify two major challenges for future molecular AI systems. Visual information alone does not guarantee reliable molecular instantiation, while Advanced bioactivity reasoning remains difficult due to the need to integrate structural understanding with empirical chemical knowledge. Addressing these challenges requires models that can jointly ground visual, textual, and symbolic chemical information and execute chemically precise transformations. Such grounded scientific foundation models will be essential for reliable AI-assisted molecular discovery and design.

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Filter step	Remaining	Removed
Raw MolParser-7M <code>sft_real</code>	91,166	—
+ Has <code><a></code> tag (pre-filter)	46,727	44,439
+ Standard R-group token only	19,282	27,445
+ No non-standard tokens	14,476	4,806
+ R-group count ≤ 10	14,394	82
+ RDKit-parseable backbone	14,394	0

Table 5: Sequential filtering statistics. The pre-filter (row 2) is applied during dataset download and retains only samples containing at least one `<a>` annotation tag; the remaining four steps correspond to the criteria listed above.

Valko, A. T.; and Johnson, A. P. 2009. CLiDE Pro: the latest generation of CLiDE, a tool for optical chemical structure recognition. *Journal of chemical information and modeling*, 49(4): 780–787.

Wang, X.; Zhang, Y.; Zhang, X.; Yu, L.; Lin, X.; Jiang, J.; Ma, B.; and Yu, K. 2024. PatentAgent: Intelligent Agent for Automated Pharmaceutical Patent Analysis. *arXiv:2410.21312*.

Zhuang, Y.; Shen, D.; and Sun, Y. 2025. MolEditRL: Structure-Preserving Molecular Editing via Discrete Diffusion and Reinforcement Learning. *arXiv preprint arXiv:2505.20131*.

A Markush Filtering Criteria

We filter the 91,166 samples from MolParser-7M `sft_real` according to four criteria, applied sequentially. Table 5 reports the number of samples remaining after each step.

- Presence of R-group tokens:** the E-SMILES must contain at least one standard placeholder (R, R[1]–R[9], X, Y, Z).
- Clean token set:** samples containing non-standard tokens such as `<dum>`, `<r>`, or `<c>` are excluded.
- R-group count:** samples with more than 10 R-group positions are excluded for complexity control.
- RDKit validity:** the backbone SMILES (with R-groups replaced by wildcard atoms) must be parseable by RDKit.

This yields 14,394 clean samples as the core source, from which task construction generates 55,982 unique editing operations across 12,962 Markush scaffolds. Multi-scaffold structures — where the backbone SMILES contains disconnected components—are not used, as they introduce ambiguity in VQA construction: it is undefined which scaffold component the instruction refers to.

B Substituent Pool Design

All substituents are drawn from 22 structurally complex, ring-containing groups (e.g., 4-chlorophenyl, 2-pyridyl, phenylsulfonyl), organised into seven similarity clusters (Table 6). To prevent chemically invalid products, we define three context-aware sub-pools: a **halogen-position pool** (7 halogenated aryl/benzyl groups) for X/Y/Z placeholders, an

N-substituent pool (16 groups) for nitrogen attachment points, and an **O-substituent pool** (10 groups) for oxygen attachment points—each a strict subset of the general pool.

C Instruction Styles

Table 7 shows the five instruction templates used in R-GROUNDBENCH. Three direct-name templates (Direct, Conversational, and Result-oriented) are used for Easy and Medium splits, as they provide explicit substituent names. Two chemically grounded templates (Relational and Descriptive) are reserved for Hard splits. Instead of directly specifying the substituent name, these templates require models to infer the intended R-group from structural relations or chemical descriptions.

The two Hard templates target complementary forms of R-group grounding. The **Relational** template does not provide an explicit substituent name or an isolated description of the target group; instead, it identifies the correct candidate through comparisons among the four R-group fragments (e.g., the greatest number of halogen atoms or the only nitrogen-containing ring). Such instructions require models to reason over the candidate set rather than rely on direct name matching, and they are generated only when the specified relation is uniquely satisfied by one candidate.

The **Chemically descriptive** template replaces explicit substituent names with chemically meaningful descriptions. It may introduce shared properties among multiple candidates (e.g., “a para-halogenated phenyl ring”) and provide additional structural constraints to identify the target (e.g., “the one whose halogen is fluorine rather than chlorine”). Therefore, successful prediction requires mapping chemical descriptions to the corresponding molecular structure instead of matching surface-level names.

For scaffolds with multiple R-positions, one clause is generated per position and the clauses are concatenated (“At R₁, ...; and at R₂, ...”). For every Hard question we verify that exactly one of the four candidates satisfies all constraints; questions failing this uniqueness check are discarded.

D Dataset Statistics and Diversity Analysis

Physicochemical property distributions. Figure 4 shows that the 55,982 edited molecules exhibit drug-like profiles: MW mean 428 Da with 70% in 300–700 Da; heavy atom count, LogP, ring count (1–3 dominant), TPSA ($<100 \text{ \AA}^2$), and H-bond donors all follow log-normal distributions consistent with oral drug space.

Scaffold diversity. Murcko scaffold analysis (Figure 5) reveals 23,822 unique scaffolds. Singletons account for 82.1% ($n = 19,569$); the top 3,618 scaffolds cover only 50% of molecules, and even the top 8,000 reach only 62%—confirming no chemotype bias.

ChEMBL Advanced data quality. The 308 SAR clusters span 13 target classes (Kinase 164, GPCR 63, Enzyme 47, Nuclear Receptor 34; Figure 6). The 2,387 pIC_{50} values follow a near-normal distribution ($\mu = 6.92$, $\sigma = 1.19$), covering the full micromolar-to-sub-nanomolar potency range relevant to drug discovery.

Group	Size	Members
G0	5	phenyl, 4-F-phenyl, 4-Cl-phenyl, 4-Me-phenyl, 4-OMe-phenyl
G1	3	3-chlorophenyl, 3,4-dichlorophenyl, 4-(trifluoromethyl)phenyl
G2	3	2-pyridyl, 3-pyridyl, 4-pyridyl
G3	3	cyclohexyl, cyclopentyl, cyclopropyl
G4	3	benzyl, 4-fluorobenzyl, 4-chlorobenzyl
G5	3	benzoyl, phenylacetyl, 4-fluorobenzoyl
G6	2	phenylsulfonyl, 4-methylsulfonylphenyl

Table 6: Seven similarity clusters of the 22-substituent pool. Within-group members share a common scaffold but differ in ring size, substitution pattern, or heteroatom position, providing challenging fine-grained alternatives for Hard splits.

Style	Split	Example
Direct	Easy/Med	Replace R_1 with 4-methylphenyl.
Conversational	Easy/Med	Can you change R_1 to 4-methylphenyl?
Result-oriented	Easy/Med	I need the R_1 position to be 4-methylphenyl.
Relational	Hard	Among the four candidates, select the one whose R_1 substituent contains the greatest number of halogen atoms.
Chemically descriptive	Hard	Replace R_1 with a para-halogenated phenyl ring, specifically the one whose halogen is fluorine rather than chlorine.

Table 7: Five instruction templates used in R-GROUNDBENCH. The upper three are direct-name styles deployed in Easy and Medium splits. The lower two are Hard templates that require additional grounding beyond explicit substituent names: **Relational** identifies the target through relations among candidates, while **Chemically descriptive** specifies the target through structural descriptions rather than direct naming.

E From VQA to Generation: A Cross-Track Performance Cliff

Figure 7 visualizes the cross-track degradation pattern across representative s2s models, complementing the quantitative analysis.

F NOTA Robustness

G Advanced Property Reasoning

H Evaluation Scope of Chemical-domain Models

Chemical-domain models are developed for different molecular understanding objectives, and therefore their applicable evaluation settings differ from those of general-purpose LLMs and VLMs.

Chemical VLMs. ChemVLM-8B and MolVL-7B are evaluated on the VQA track to assess molecular recognition

Statistic	Value
Source subset	MolParser-7M <code>sft_real</code>
Raw patent/lit. mols	91,166
Markush-containing	46,727
Clean samples used	14,394
Unique Markush scaffolds	12,962
Unique editing operations	55,982
Total instruction records	56,500
Basic VQA questions	940 per split
Advanced VQA questions	634 per split
NOTA questions	≈20% of Easy/Med Basic
Generation questions	500 easy + 500 hard
Input modalities	4 (i2i / i2s / s2i / s2s)
ChEMBL SAR clusters	308
Bioactivity data points	2,387 (mean pIC_{50} = 6.92)
Biological target classes	13

Table 8: R-GROUNDBENCH dataset statistics.

and candidate selection. Their training objectives primarily focus on molecular representation alignment and structure–text understanding rather than explicit instruction-grounded molecular editing. Advanced VQA introduces additional requirements, including physicochemical property estimation, structure–activity relationship (SAR) understanding, and biological activity interpretation, which are beyond the primary objectives of existing chemical VLM pretraining. Therefore, we focus their evaluation on Basic VQA and discuss Advanced VQA performance separately in Appendix.

Markush-specific model. MarkushGrapher-2 is specifically designed for Markush structure reconstruction and molecular generation, rather than general-purpose molecular question answering or instruction following. We therefore evaluate it on the Generation track, which partially overlaps with its original objective of converting Markush representations into instantiated molecular structures. The reported evaluation uses molecular images as input, consistent with its original image-based Markush reconstruction setting.

However, our Generation track additionally requires instruction-grounded R-group resolution, where models must interpret natural-language editing instructions and perform the corresponding substituent substitution. This capability is not explicitly targeted by MarkushGrapher-2. VQA evaluation is not directly comparable because it provides a predefined candidate set and measures candidate-level

Molecular Property Distributions of R-GroundBench (Edit Dataset)

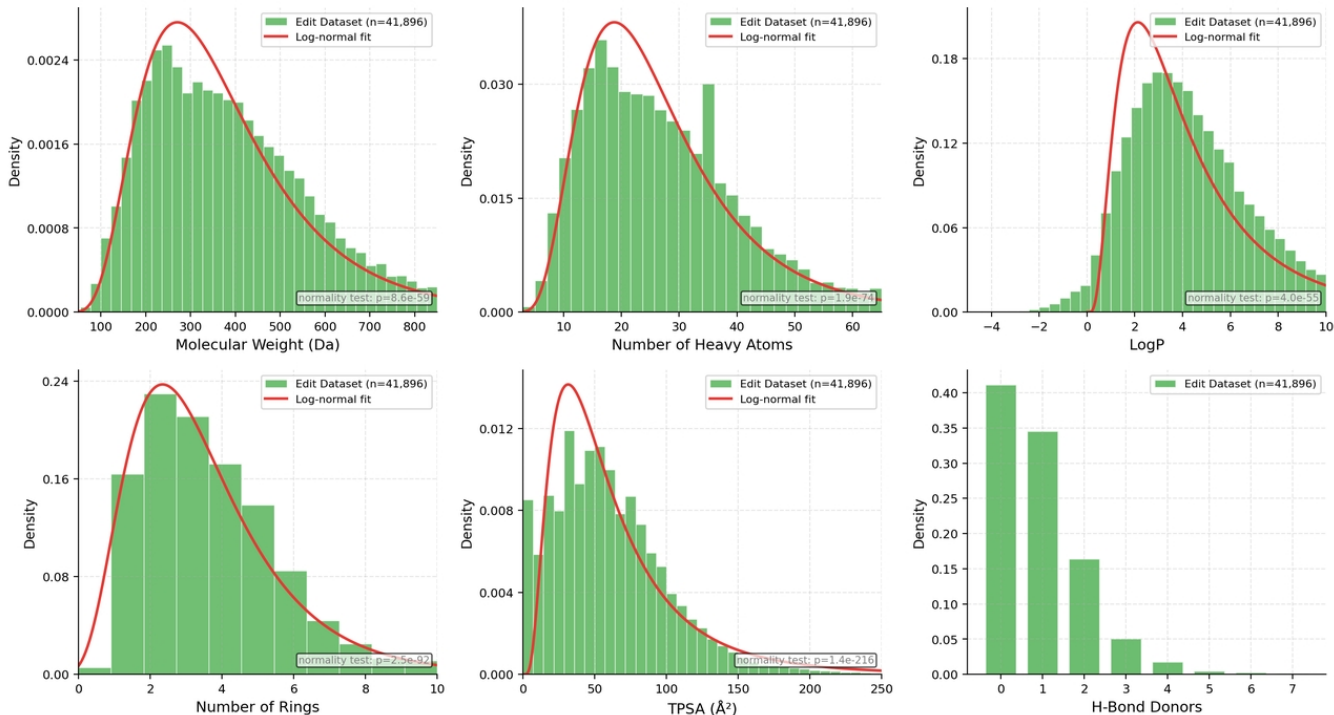


Figure 4: Physicochemical property distributions of edited molecules ($n = 55,982$). Green histograms with log-normal fits (red).

discrimination, whereas Generation evaluates autonomous molecular instantiation without candidate constraints.

I Quantitative Analysis of Shortcut Control

To quantify whether the difficulty splits successfully control similarity-based shortcuts, we define a per-question *similarity gap*:

$$\Delta_{\text{sim}} = \text{Tanimoto}(c, s) - \frac{1}{3} \sum_w \text{Tanimoto}(w, s), \quad (1)$$

where c is the correct option, w ranges over the three distractors, and s is the Markush scaffold using ECFP4 fingerprints (2048-bit). A positive Δ_{sim} indicates that the correct option has a scaffold-similarity advantage, creating a potential shortcut, whereas values close to zero indicate that this structural cue is removed.

By construction, Easy-Basic yields a mean $\Delta_{\text{sim}} = +0.33$, while Medium-Basic and Hard-Basic both yield approximately 0.00, showing that similarity-based shortcuts are largely restricted to Easy-Basic. Table 11 reports accuracy across Δ_{sim} bins for 8 VLMs under the img-smi setting. Easy-Basic shows a clear downward trend as similarity decreases (99.3% at $\Delta_{\text{sim}} \geq 0.50$ to 94.3% at $[-0.05, 0)$), indicating that scaffold similarity provides useful shortcuts in this setting. In contrast, Hard-Basic remains nearly unchanged across similarity bins, suggesting that its difficulty mainly arises from instruction-grounded structural reasoning rather than distractor similarity.

Bootstrap significance of difficulty gaps. To verify that the Easy-to-Hard accuracy drop is not attributable to sampling variability, we apply bootstrap resampling ($n=10,000$ iterations, seed 42) to each split independently. Mean accuracies with 95% confidence intervals are: Easy-Basic 98.7% [98.22, 99.08], Medium-Basic 93.7% [92.76, 94.62], and Hard-Basic 62.4% [59.52, 65.11]. The Easy-to-Hard gap $\Delta_{\text{EH}} = 36.3$ percentage points (95% CI [33.51, 39.10], $p < 0.001$, one-tailed bootstrap test), confirming that the performance cliff is statistically significant. The Easy-to-Medium gap is 5.0 percentage points (95% CI [3.96, 5.98], $p < 0.001$), and the Medium-to-Hard gap is 31.4 percentage points (95% CI [28.40, 34.27], $p < 0.001$).

Correlation between similarity gap and accuracy. To further examine whether scaffold similarity predicts model accuracy, we compute the Spearman correlation ρ between per-question Δ_{sim} and mean accuracy across 8 VLMs, with bootstrap 95% confidence intervals ($n=10,000$). We focus on VLMs because this analysis targets visual candidate recognition under image-based input settings. The correlations are: Easy-Basic $\rho = +0.138$ ($p < 0.001$, 95% CI [0.06, 0.21]), Medium-Basic $\rho = +0.025$ ($p = 0.501$, 95% CI $[-0.05, 0.10]$), and Hard-Basic $\rho = -0.008$ ($p = 0.815$, 95% CI $[-0.07, 0.06]$). The significant positive correlation on Easy-Basic indicates that questions with stronger scaffold-similarity advantages are easier for models. The near-zero, non-significant correlations on Medium- and Hard-Basic

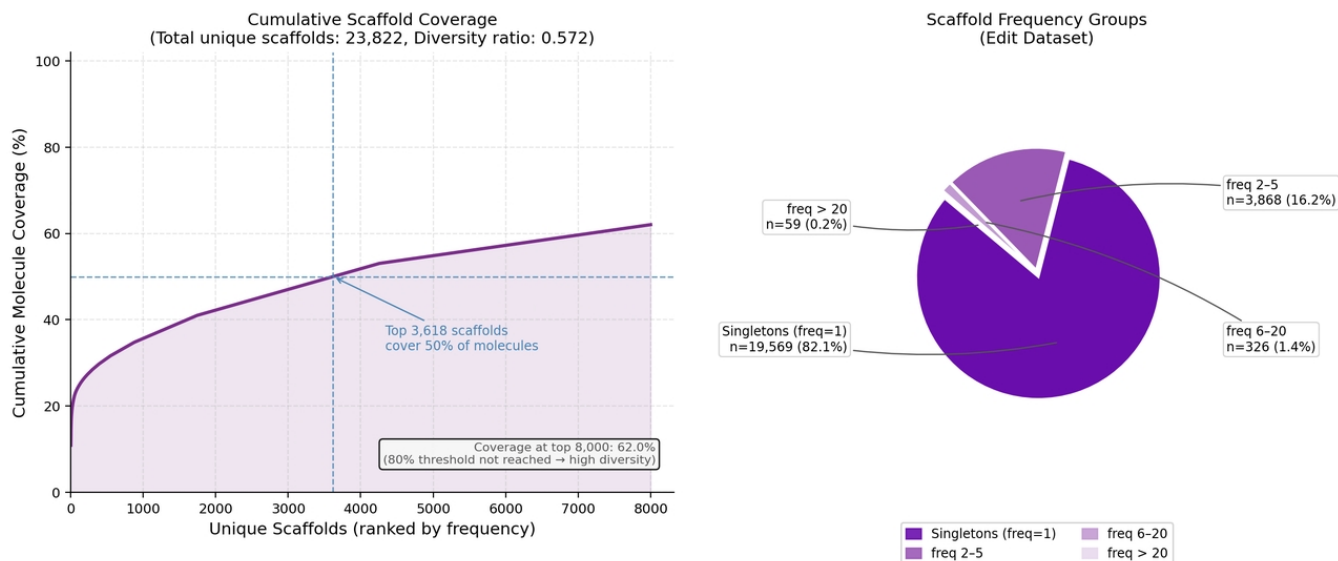


Figure 5: Scaffold diversity. **Left:** cumulative coverage curve (23,822 unique scaffolds). **Right:** frequency group distribution (82.1% singletons).

	NOTA is correct		
Model	Struct. err	Non-exist R	NOTA as distr.
VLMs (img-smi)			
Qwen3-VL-8B	61.0	75.8	73.3
Qwen3-VL-32B	88.6	97.0	72.1
Claude Sonnet 4.6	92.9	<u>98.0</u>	80.2
GPT-4o	90.0	94.9	72.1
GPT-4.1	76.2	80.8	98.8
GPT-5.5	98.1	100.0	76.7
Gemini 2.5 Flash	<u>93.8</u>	100.0	67.4
Llama 4 Maverick	40.4	50.5	<u>95.3</u>
Average	80.1	86.6	79.5
LLMs (smi-smi)			
DeepSeek-V4-Pro	80.6	88.5	63.6
DeepSeek-R1	<u>98.4</u>	99.0	28.8
LLaMA-3.3-70B	54.3	84.8	93.0
o3	100.0	<u>97.1</u>	58.3
Qwen3-Max	92.3	<u>95.9</u>	<u>90.5</u>
Average	85.1	93.1	66.8

Table 9: NOTA robustness (%; img-smi for VLMs; smi-smi for LLMs). **Red bold:** best; underline: second best.

confirm that distractor similarity no longer predicts accuracy after shortcut control.

J Evaluation Prompts

All models are evaluated under a standardized zero-shot protocol with temperature set to 0. We describe the system prompt and user prompt templates for each evaluation track below. The VQA track covers all four input-output modalities (img-img, img-smi, smi-img, smi-smi); the Generation track covers smi-smi and img-smi only.

Model	Physchem	Bioactivity	Indication
<i>VLMs (img-smi, Hard)</i>			
Qwen3-VL-8B	85.5	26.3	83.7
Qwen3-VL-32B	89.3	26.7	<u>88.4</u>
Claude Sonnet 4.6	66.7	23.9	82.6
GPT-4o	86.2	29.5	82.1
GPT-4.1	84.9	26.0	88.9
GPT-5.5	78.6	27.4	86.8
Gemini 2.5 Flash	<u>88.7</u>	<u>27.4</u>	83.7
Llama 4 Maverick	79.2	24.9	80.5
Average	82.4	26.5	84.6
<i>LLMs (smi-smi, Hard)</i>			
DeepSeek-V4-Pro	70.7	28.7	87.9
DeepSeek-R1	51.2	33.0	51.7
LLaMA-3.3-70B	<u>78.6</u>	24.9	43.2
o3	58.1	24.3	<u>63.0</u>
Qwen3-Max	83.0	<u>27.0</u>	62.6
Average	68.3	27.6	61.7

Table 10: Advanced property reasoning accuracy (%; Hard split) by property type. **Red bold:** best; underline: second best.

J.1 VQA Track

System prompt (all VQA settings).

You are an expert chemist answering multiple choice questions. YOUR ONLY OUTPUT MUST BE A SINGLE LETTER: A, B, C, or D. DO NOT write any explanation, reasoning, analysis, or punctuation. DO NOT start your response with words. ONLY output one of these four characters: A B C D

ChEMBL SAR Data Quality for Advanced VQA Tasks

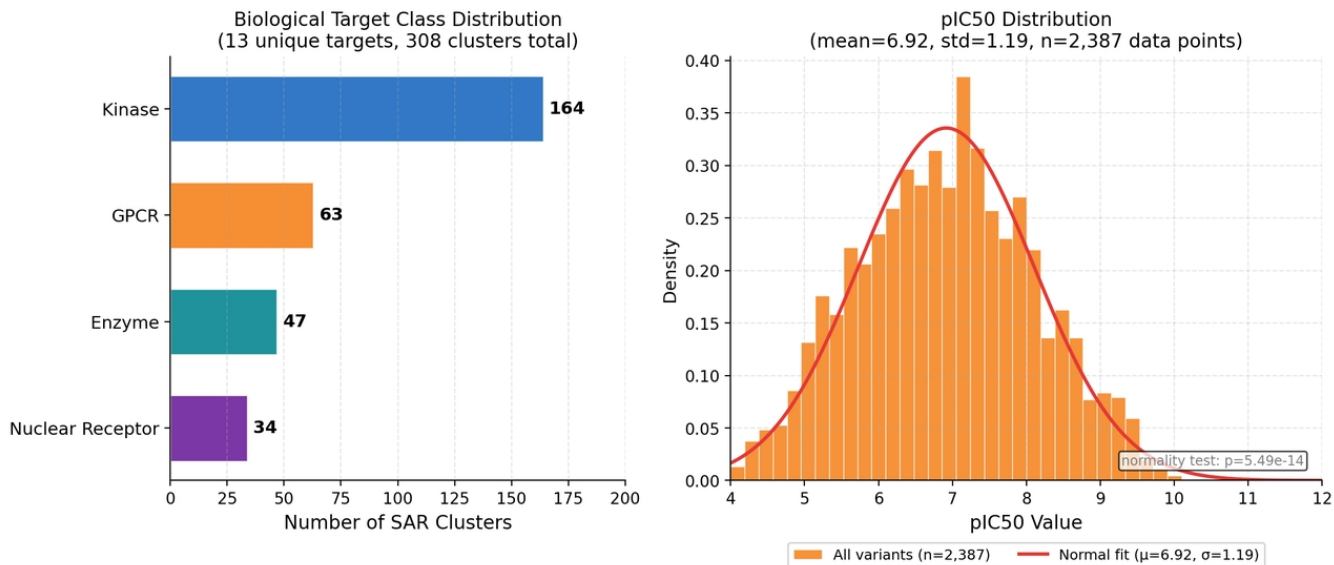


Figure 6: ChEMBL SAR data quality. **Left:** target class distribution (308 clusters, 13 types). **Right:** pIC₅₀ distribution ($\mu = 6.92$, $\sigma = 1.19$, $n = 2,387$).

Gap Δ_{sim}	Easy	Medium	Hard
< -0.05	—	91.9	60.3
$[-0.05, 0.00)$	94.3	93.3	63.4
$[0.00, 0.05)$	95.7	93.6	61.2
$[0.05, 0.10)$	98.4	98.4	62.1
$[0.10, 0.20)$	98.0	—	—
$[0.20, 0.50)$	98.6	—	—
≥ 0.50	99.3	—	—

Table 11: Accuracy (%) by similarity gap Δ_{sim} (Eq. 1), averaged over 8 VLMs under the img-smi setting. NOTA questions are excluded. The lower rows correspond to Easy-Basic samples with larger scaffold-similarity advantages, which are absent in Medium- and Hard-Basic due to controlled distractor construction. “—”: $n < 10$.

User prompt — Basic, smi-smi / img-smi. The Markush structure is provided either as an image (img-smi) or as an E-SMILES string (smi-smi); the four candidates are always provided as SMILES strings.

```
{Markush image} [img-smi only]
Markush structure (SMILES): {E-SMILES}
[smi-smi only]
Instruction: {instruction}
{question}
Candidate molecules (SMILES):
A: {SMILES_A}
B: {SMILES_B}
C: {SMILES_C}
D: {SMILES_D}
```

[NOTA only] Note: Option D is ‘None of the above’. Select D if no option is chemically valid or correct.

Your answer (single letter only, no explanation):

User prompt — Basic, smi-img / img-img. The four candidates are provided as molecule images appended after the text.

```
{Markush image} [img-img only]
Markush structure (SMILES): {E-SMILES}
[smi-img only]
Instruction: {instruction}
{question}
The four candidate molecules are shown below as images (A, B, C, D).
[NOTA only] Note: Option D is ‘None of the above’. Select D if no option is chemically valid or correct.
{image_A} {image_B} {image_C} {image_D}
Your answer (single letter only, no explanation):
```

For any candidate whose molecule is “None of the above” under an image-candidate setting, the corresponding image slot is replaced by the text line Option {label}: None of the above.

User prompt — Advanced. Advanced tasks omit the Instruction line and replace it with a property-based {question}. Indication/Mechanism questions present the four options as *R-groups* (labelled “Candidate R-groups”), whereas Physicochemical and Bioactivity questions present

full *candidate molecules* (labelled “Candidate molecules”). As in the Basic task, candidates are rendered as SMILES in the smi-smi / img-smi settings and as appended images in the img-img / smi-img settings; the Markush input line additionally notes “with an R-group position”. Advanced tasks contain no NOTA variant.

```
{Markush image} [img-smi only]
Markush structure (SMILES) with an
R-group position: {E-SMILES} [smi-smi
only]
{property question}
Candidate molecules (SMILES):
[Physicochemical / Bioactivity]
Candidate R-groups (SMILES):
[Indication/Mechanism]
A: {SMILES_A}
B: {SMILES_B}
C: {SMILES_C}
D: {SMILES_D}

Your answer (single letter only, no
explanation):
```

In the img-img / smi-img settings, the four option lines above are instead appended as images after the question, exactly as in the Basic image-candidate prompt.

J.2 Generation Track

The Generation track is evaluated under two input modalities only, smi-smi (E-SMILES in, SMILES out) and img-smi (image in, SMILES out); there are no candidate options and hence no NOTA variant. Both modalities share the same system prompt and differ only in how the Markush structure is presented.

System prompt.

```
You are an expert chemist. Given a
Markush structure and an R-group
substitution instruction, generate
the SMILES string of the resulting
molecule. Return ONLY the SMILES string
enclosed in <smiles> and </smiles>
tags. Do not include any explanation,
reasoning, or additional text. Example
format: <smiles>CCO</smiles>
```

User prompt — smi-smi (E-SMILES input). The Markush structure is provided as an E-SMILES string.¹

```
Markush structure (SMILES):
{E-SMILES}

Instruction: {instruction}

Generate the SMILES of the resulting
molecule after applying this
instruction. Return ONLY the SMILES
enclosed in <smiles></smiles> tags.
```

¹The field is labelled “(SMILES)” in the prompt for brevity, but the value passed is the full E-SMILES string with R-group position annotations.

User prompt — img-smi (image input). The Markush structure is provided as a rendered image, prepended before the text content.

```
{Markush image}

The image above shows a Markush
structure with R-group position(s).

Instruction: {instruction}

Generate the SMILES of the resulting
molecule after applying this
instruction. Return ONLY the SMILES
enclosed in <smiles></smiles> tags.
```

Output parsing. We extract the predicted SMILES from the model response in two stages. First, we search for the <smiles>...</smiles> tag pattern (case-insensitive, spanning newlines) and, if found, take its enclosed content. If no tag is present, we fall back to the last non-empty line of the response, accepting it as a SMILES only if it contains no whitespace and is shorter than 300 characters; otherwise the prediction is treated as empty (and thus invalid). The extracted string is then canonicalized via RDKit before computing Exact Match, Scaffold Match, and Tanimoto Similarity.

K Limitations

Patent-level complexity. Although R-GroundBench is constructed from real patent and literature Markush structures, the benchmark focuses on well-defined R-group grounding instances with controlled instructions and structured evaluation settings. Real patent documents may contain more complex R-group definitions involving long-range textual dependencies, conditional clauses, and heterogeneous document layouts. Nevertheless, the current benchmark follows common Markush description patterns in patent claims and already reveals substantial grounding failures in existing models. Extending evaluation toward full patent-level retrieval and reasoning remains an important future direction.

Multiple-choice and generation scope. The VQA track uses controlled candidate selection to isolate R-group grounding ability, although multiple-choice settings may still permit limited structure-level shortcuts. The Generation track evaluates end-to-end molecular instantiation through free-form SMILES generation, but it currently focuses on instruction-guided R-group substitutions rather than unconstrained molecular discovery. Future benchmarks could incorporate broader document-level reasoning and more open-ended molecular design scenarios.

L The Use of Large Language Models

We used large language models (GPT-4o and Claude Sonnet 4.6) only as general-purpose writing assistants to refine the clarity and fluency of the manuscript, such as improving grammar, style, and consistency of academic tone. All scientific ideas, methodological designs (including the R-GroundBench framework, algorithms, and experiments), data collection, and analysis were conceived and executed

entirely by the authors. Any LLM-generated or suggested text was critically reviewed and edited to ensure accuracy and faithfulness to our original contributions.

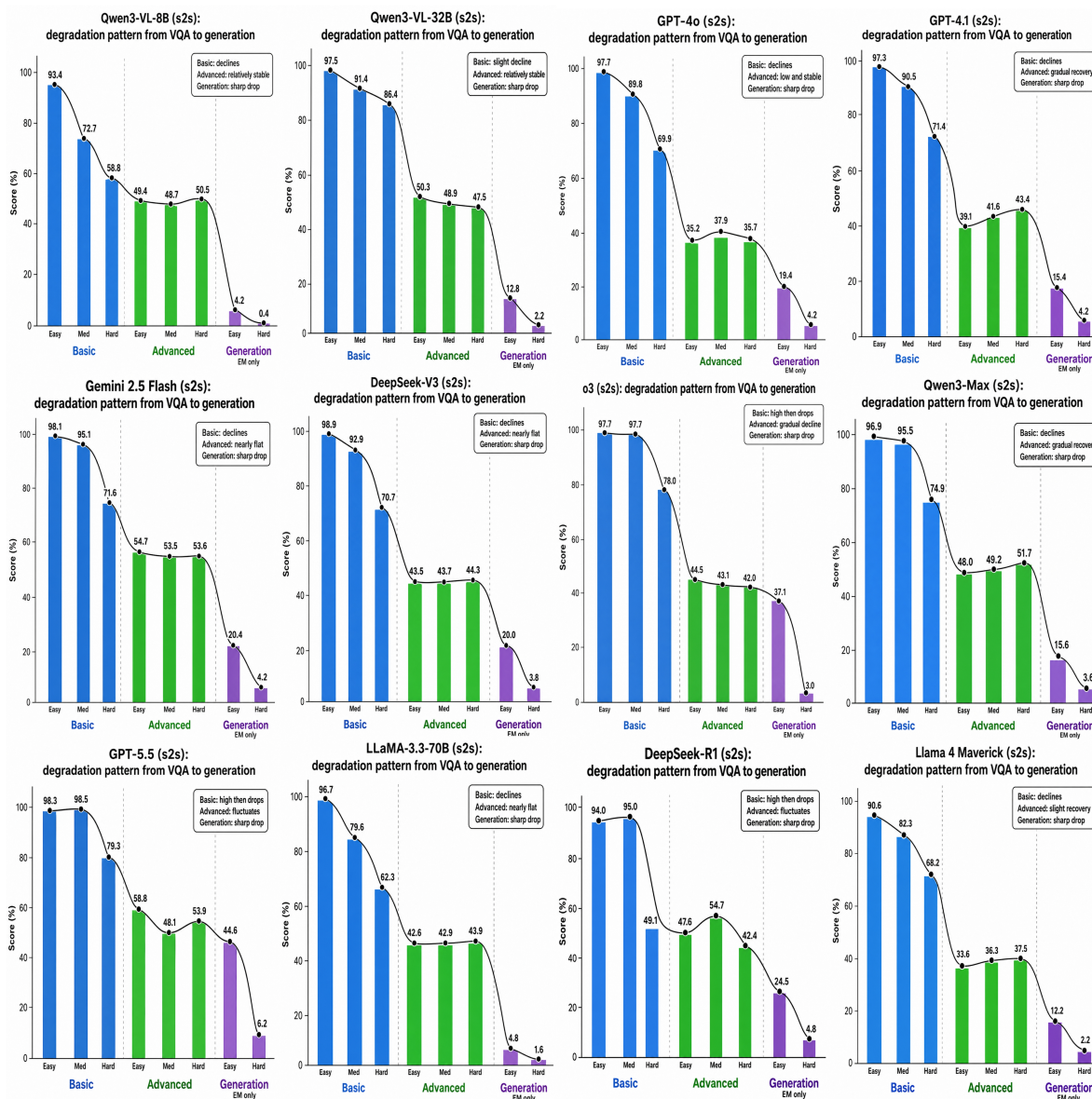


Figure 7: Cross-track degradation from VQA to Generation under the s2s modality. For each model, blue bars show Basic VQA accuracy, green bars show Advanced VQA accuracy, and purple bars show Generation Exact Match (EM). The black curve connects all scores to highlight the diagnostic performance cliff: models that perform well on VQA, especially Basic Easy/Medium splits, collapse when required to generate the edited molecule directly. This pattern indicates that multiple-choice success often reflects shortcut exploitation rather than robust atomic-level R-group grounding.